

ECE 364: Assignment #2

1. (10 pts)
 - (a) (5 pts) Compute the Mahalanobis distance between the data instances (10,20,30) and (15,15,28) if the covariance matrix is $2I$, where I is the identity matrix.
 - (b) (5 pts) What is the Minkowski distance, with parameter $p = 4$, between (3,5,7,9) and (7,9,3,5)?
2. (10 pts) A data analyst building a k -nearest neighbor model for a continuous prediction (i.e., regression) problem is considering appropriate values for k .
 - (a) (5 pts) Initially, the analyst uses a simple average of the target feature values for the k nearest neighbors in order to make a prediction. After experimenting with values in the 1-10 range, it occurs to the analyst that setting the value of k to the total number of data instances in the training set may yield very good results. Do you think this is true? Explain.
 - (b) (5 pts) If the analyst was using a distance-weighted average rather than a simple average to make predictions, would it make the analyst's idea any more useful? Explain.
3. (10 pts)
 - (a) (4 pts) Consider the following five data instances given as $((x_i, y_i), c_i)$ values, where x_i and y_i are values of the two descriptive features (positive integers) and c_i is a binary class label: $((1,1), -1)$, $((1,7), +1)$, $((3,3), +1)$, $((5,4), -1)$, $((2,5), -1)$. Classify a query at coordinates (3,6) using a 3-NN classifier based on Manhattan distance. Your answer should either be +1 or -1.
 - (b) (2 pts) True or False: The decision boundary associated with a 1-NN classifier defined by the above five data instances consists of a sequence of line segments that are each parallel to exactly one of the two axes associated with the two descriptive features, x and y . Explain.
 - (c) (2 pts) True or False: The predictions of a general k -NN classifier that uses Euclidean distance may change if we multiply each data instance's descriptive feature values by 0.5. Explain.
 - (d) (2 pts) True or False: As the value of k used in a k -NN classifier is incrementally increased from 1 to n , where n denotes the total number of data instances, the classification accuracy on the training dataset will always increase. Explain.

(10 pts) Prove that the Russel-Rao similarity index is not a metric.

4. (20 pts) Coding project

In this project, you will train classifiers based on the k -nearest neighbors algorithm to predict whether or not a female patient has diabetes. The dataset consists of 767 patients with the following eight numerical descriptive features each:

- Number of pregnancies

- Glucose
- Diastolic blood Pressure
- Skin thickness
- Insulin
- BMI
- Pedigree
- Age

The target label is a binary variable indicating whether or not a patient has diabetes.

The data will be divided and preprocessed into a training set and a validation set. You will start with the default kNN classifier in the Scikit-learn library and evaluate the effect of varying the number of neighbors. You will also train kNN classifiers that use distance weighting and different power parameters for the Minkowski metric.

Please see the Jupyter notebook for more details.

GitHub repository for ECE364 coding projects: <https://github.com/jha-lab/ECE364.2026>